

# Measuring Segregation Using Convex Functions

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## Outline

Broad motivation for the article

- Folks in the math community are turning to quantitative measures of segregation [DMW23, TBK<sup>+</sup>25, HSS25].
- There's actually a pretty interesting and rich literature in quantitative sociology on this set of topics [Rea, RF02], going back all the way at least to [Sim49].
- MAUP is a thing [Ope84], and we'd like measures that behave in predictable ways as we change analytical scale.
- If you start engaging with these ideas systematically and mathematically, then you come naturally to convexity and Bregman divergences, which are what I am here to sell you today.

Desirable properties of segregation measures.

blah blah blah

More blah, more blah

das

Figure 1 is a simple checkerboard pattern.

- What properties to have? Pielou criteria [Pie77], various axioms from [RF02], etc.

This document can serve as a template for your article. Here are a few things to keep in mind. Articles should be understandable by 2nd year math grad students, and should be in the style of a good colloquium talk where

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[Cho25] [Cho17]

[BMDG05]

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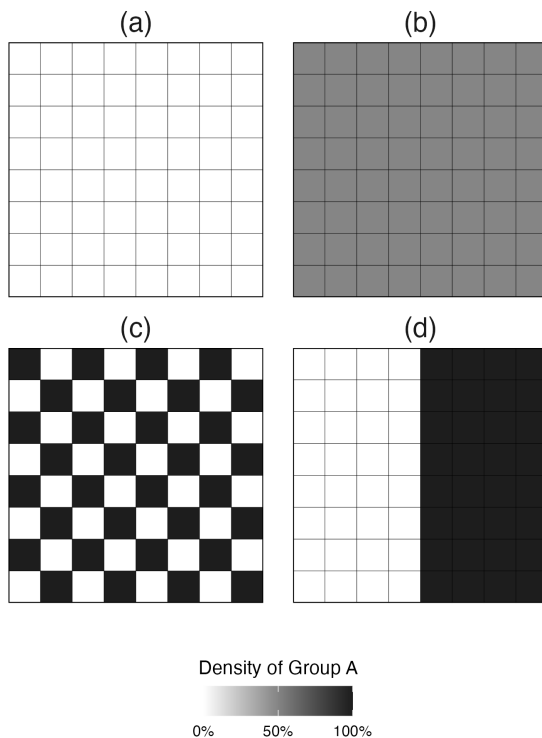


Figure 1: A simple checkerboard pattern.

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  JOURNAL = {Ann. of Math. (2)},
  FJOURNAL = {Annals of Mathematics.
    Second Series},
  VOLUME = {41},
  YEAR = {1940},
  PAGES = {1--7},
  ISSN = {0003-486X},
  MRCLASS = {09.1X},
  MRNUMBER = {0001219},
  MRREVIEWER = {R. Brauer},
  DOI = {10.2307/1968817},
  URL = {http://doi-org/10.2307/1968817},
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## References

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